

Title

Implementation Pathways and Health Service Utilization Patterns in Adults With Inflammatory Bowel Disease Receiving Digital-Facilitated Self-Management Support

Abstract

Understanding health service utilization and implementation factors informs dissemination strategy. Among the 780 trial participants, secondary analyses examined patterns of intervention engagement, changes in outpatient service use, medication escalation patterns, and work-related impacts. Participants in the digital-intervention arm attended fewer gastroenterology outpatient visits (median 2.1 vs. 2.9 visits during follow-up; $p=0.02$) and reported greater work productivity (mean productivity loss 18% vs. 28%; $p=0.001$). Medication changes (escalation or de-escalation) were similar between arms (43% vs. 45%), suggesting the intervention did not substitute for appropriate medical management. Engagement with digital modules was associated with superior cost-effectiveness (participants completing >60% of content showed greater cost savings). Qualitative data revealed participant preference for symptom-focused support complementing medical care. Unplanned healthcare visits and emergency department use showed trending reductions in the intervention group (27% vs. 34%), suggesting potential additional benefits in healthcare access.